KENDRIYA VIDYALAYA SANGATHAN BHUBANESWAR REGION

PRE- BOARD EXAMINATION (2024-25)
CLASS – X
SUBJECT -MATHEMATICS STANDARD (041)

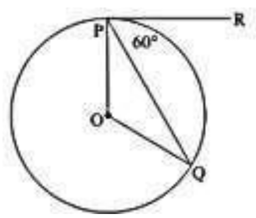
TIME: 3 Hrs **M:M- 80**

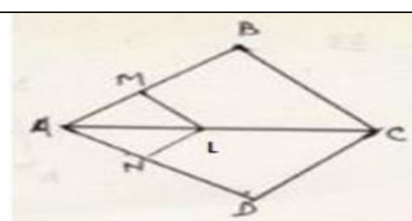
GENERAL INSTRUCTIONS:

1. The question paper consists of 38 questions. All questions are compulsory.
2. This question paper is divided into five sections - A, B, C, D and E.
3. In Section A, Questions 1 to 18 are Multiple Choice Questions (MCQs) and question number 19 & 20 are Assertion – Reason Based questions of 1 mark each.
4. In Section B, Question Number 21 to 25 are Very Short Answer (VSA) type questions carrying 2 marks each.
5. In Section C, Question Number 26 to 31 are Short Answer (SA) type questions carrying 6 questions of 3 marks each.
6. In Section D, Question Number 32 to 35 are Long Answer (LA) type questions carrying 5 marks each.
7. In Section E, Question Number 36 to 38 are Case Study Based Questions carrying 4 marks each. Internal Choice is provided in 2 marks questions in each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section – B, 2 questions in Section – C, 2 questions in Section – D, and 3 questions in Section – E,
9. Draw a neat figure wherever required. Take $\pi = 22/7$ wherever required if not stated.

S.No	QUESTION
	SECTION A [1X20 =20]
1	Which are the zeroes of $p(x) = 6x^2 – 7x – 3$ (a) 5, –2 (b) –5, 2 (c) –5, –2 (d) none of these
2	The ratio in which the X-axis divides the line segment joining A(3, 6) and B (12, -3) is: (a) 2:1 (b)1:2 (c) -2:1 (d) 1: -2
3	The nth term of an A.P. 5, 2, -1, -4, -7 ... is (a) 2n + 5 (b) 2n - 5 (c) 8 - 3n (d) 3n – 8

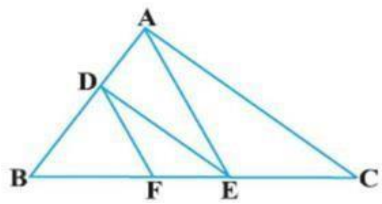
4	The total surface area of a solid hemisphere of radius 7 cm mounted on a hollow cylinder of same radius and height 3cm is : (a) 447π cm^2 (b) 239π cm^2 (c) 175π cm^2 (d) 140π cm^2
5	If one zero of the quadratic equation $x^2 + 3x + k$ is 2, then the value of k is (a) 10 (b) -10 (c) 5 (d) -5
6	The difference of median and mode of a data is 24, then find the difference of mean and median is a.8 b. 12 c. 24 d 36
7	<table><tr><td>MARK</td><td>15-20</td><td>20-25</td><td>25-30</td><td>30-35</td><td>35-40</td><td>40-45</td></tr><tr><td>NO.OF STUDENTS</td><td>3</td><td>8</td><td>9</td><td>10</td><td>3</td><td>6</td></tr></table>Find the upper limit of the modal class a. 10 b. 30 c. 9 d.35
8	A point Q which is 13 cm away from the centre of a circle , the length of tangent PQ to the circle is 12 cm. The radius of the circle (in cm) is a)25 (b) $\sqrt{313}$ (c)5 (d)1
9	Find the value of k for which the lines $-2x + 5y = 15$ and $-10x - ky = 75$ have infinite number of solutions. (a). 5 (b) -5 (c) 25 (d) -25
10	The distance between the points (a, b) and (-a, -b) is: ________ (a). 2a (b) $2\sqrt{(a^2 + b^2)}$ (c) 2b (d) $2\sqrt{(a^2 - b^2)}$
11	A sphere and a cube have equal surface areas. The ratio of the volume of the sphere to that of cube is (a) $\sqrt{\pi} : \sqrt{6}$ b) $\sqrt{6} : \sqrt{\pi}$ (c) $\sqrt{\pi} : \sqrt{3}$ (d) $\sqrt{3} : \sqrt{\pi}$
12	Find the value of Cos 60° sin 30° + Sin 60°cos 30° (a) -1 (b) 1 (c) 2 (d) 3
13	A card is randomly selected from a pack of 52 cards. What is the probability of getting neither a red king nor a black king a) 12/13 b) 7/13 c) 12/26 d) none of these
14	(sec A + tan A) (1 –sin A) = (a) sec A (b) sin A (c) cosec A (d) cos A
15	If P(E) = 0.23, what is the probability of 'not E'?

	a) 0.77 b) 0.023 c) 23 d) 0.0023
16	If radii of two concentric circles are 4 cm and 5 cm ,then find the length of chord of larger circle which is tangent to the other circle. (a)8 (b) 3 (c) 6 (d) None of These
17	In the given figure, if *PR* is tangent to the circle at *P* and O is the centre of the circle and ∠QPR =60°, then ∠*POQ* =  (a) 110° (b) 100° (c) 120° (d) 90°
18	If α, β are the zeroes of the polynomial $x^2 - 16$, then αβ (α +β) (a) -4 (b) -16 (c) 0 (d) 16
19	**Statement A (Assertion**): Area swept by minute hand of a clock of length 7 cm in 15 minutes is 38.5 sq.cm . $(\pi = \frac{22}{7})$ **Statement R (Reason**) : The angle formed by minute hand of the clock in 15 minute is 90 degree. Area of sector = sector angle /360° x π (radius)^2 Which of the following is correct ? (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but rason (R) is true.
20	**Statement A (Assertion**):- The LCM of 56 and 294 is four times the larger number. **Statement R (Reason**) : The ratio of product of two numbers to its HCF is equal to the ratio of their LCM to HCF Which of the following is correct ? (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)

	(b) Both assertion (A) and reason (R) are true and reason (R) is not correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d)Assertion (A) is false but reason (R) is true.
	SECTION B [2 X 5 =10]
21.	In what ratio does the point P (-4,y) divide the line segment joining the points A (-6,10) and B (3,-8) if it lies on AB ? Hence find the value of y. **OR** Find the ratio in which the *y* -axis divides the line segment joining the points (-1, - 4) and (5, - 6). Also find the coordinates of the point of intersection.
22	If A(0,1) is equidistant from B(5,-3) and C(x,6) find value of x
23	Evaluate 2 $\tan^2 45^0 + 2\cos^2 45^0 - 2\sin^2 45^0 + \cot^2 45^0$ **OR** If cos A = 2/5, find the value of 4 + 4 $\tan^2$ A
24	Find the smallest number divisible by 60, 84 leaves remainder 2 in each case.
25	A card is drawn from a well shuffled pack of 52 cards. Find probability of getting i. A red picture card ii. A Queen or an Ace.
	SECTION C [3 X 6 =18]
26	Prove that: $(\sin A + \mathrm{cosec}\, A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$ **OR** Prove that SecA (1-SinA) (SecA+ Tan A) =1
27	In the figure LM ∥ CB and LN ∥ CD then prove that $\frac{AM}{AN} = \frac{BM}{DN}$  OR

In the figure DE ∥ AC and DF ∥ AE , Prove that

$BE^2 = BF.BC$



28 Prove that $\sqrt{5}$ is an irrational number.

29 In a circle of radius 21cm, an arc subtends an angle of 60° at the centre. Find

i) The length of the arc

ii) Area of the sector formed by the arc.(use $\pi = \frac{22}{7}$)

30. Divide 27 into two parts such that the sum of their reciprocals is $\frac{3}{20}$.

OR

A two digit number is such that product of its digits is18 and if 63 is subtracted from it the digits got interchanged. Find the number.

31 If one zero of the polynomial $(a^2 + 9)x^2 + 13x + 6a$ is reciprocal of the other .Find the value of a.

OR

Find a quadratic polynomial whose zeroes are three times the zeroes of $P(x) = 2x^2+7x+6$

SECTION D [5 X 4 = 20]

32 If the median of the distribution table given below is 28.5, find the values of x and y.

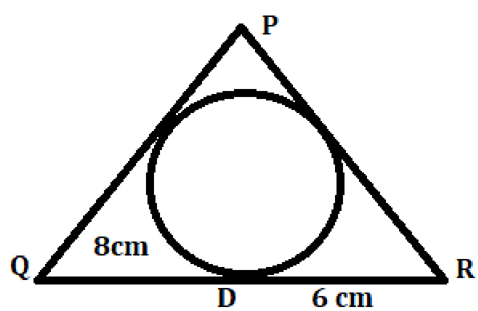
Class interval	0-10	10-20	20-30	30-40	40-50	50-60	Total
frequency	5	x	20	15	y	5	60

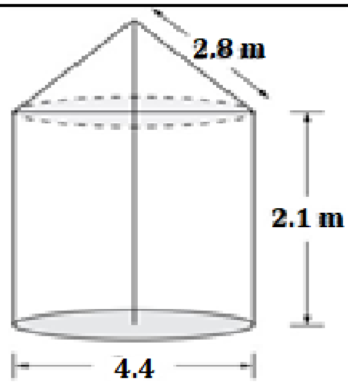
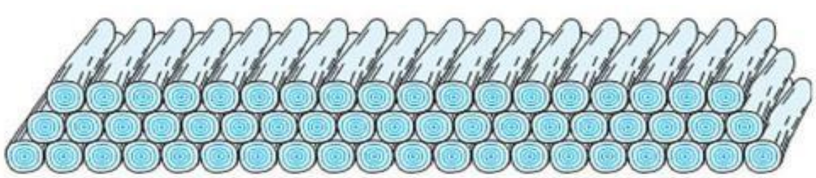
33 (a) Prove that the tangents drawn from an external point to a circle are equal in length.

(b)If a quadrilateral PQRS circumscribes a circle then prove that PQ +RS = PS + QR

OR

▲ PQR is drawn to circumscribe a circle of radius 4 cm such that the segments QD and RD of QR are of lengths 8 cm and 6 cm respectively. Find PQ and PR.



34	Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars? OR Solve the linear equations 2x+y = 6 and 2x-y+2 =0 Graphically. Also find the area of the triangular region bounded by these lines with Y axis.
35	The angle of elevation of the top of a tower from a point A on the ground is 30^0. On moving a distance of 20 meters towards the foot of the tower to a point B, the angle of elevation increases to 60^0.Find the height of the tower and the distance of the tower from the point A.($\sqrt{3}$=1.732)
SECTION E (CASE BASED) [4 X 3= 12]	
36	 For a camp of NCC different tents are to be made at the Camp site.A tent is in the shape of cylinder surmounted by a conical top of same diameter as shown in the figure. If the height and diameter of cylindrical part of the tent are 2.1 m and 4.4 m respectively and the slant height of its conical part must be 2.8 m to accommodate 5 cadets . (i) Find the vertical height of the conical part of tent. (ii) Find the volume of air inside the tent. (iii) Find the cost of canvas needed to make one such tent as the previous tents are outdated and torn. Canvas is available at the rate of Rs.500 per square meter . Or Find the weight of the tent if 1 sq. m of Canvas weigh 2 kg without any rope and essentials.
37	For supplying water to each house of a village pipes are brought to be laid to each house. For this purpose, 200 such long pipes are purchased and stacked in the following manner, 20 pipes in the bottom row, 19 in the next row, 18 in the next to it and so on.  (i) In how many rows are the 200 pipes placed? (ii) How many pipes are there in the topmost row?

Or

Find the number of pipes lying between the top and bottom row

(iii)If one such stack of 105 pipes is arranged in 6 rows, find the number of pipes at the bottom.

38 SAM is a student of class X. A mobile tower is fixed near his house. He wants to find the height of the tower using the concept of similarity for triangles that he learnt. The height of a tree near SAM'S house is 40 metre when it casts a shadow of 8-metre-long on the ground. But SAM's house casts a shadow of 15- metre-long on the same ground.



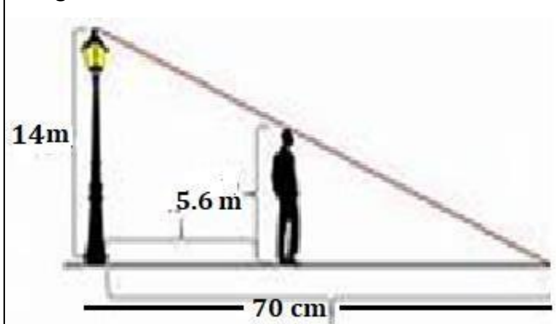
Answer the following questions based on the information given above

a. . At the same time the tower casts a shadow of 30-metre-long on the ground then find height of the tower?

b. Find the difference between the height of the tower and SAM's house when the difference of their shadows is 1.2 meter.

OR,

Two more buildings along the road standing parallel to each other and their heights are in a ratio of 3:1 and the smaller house casts a shadow of 2 m . Find the length of shadow cast by the larger house.



c. Find the difference between the height of man and his shadow if the lamp post of 14 m casts a shadow of 70cm

…………………………………………….. END………………………………………………